

SMPC Mini-Project 1

A student would like to create a 1.8V supply from a 5.0V battery. The 1.8V supply needs to supply a load resistance of 4.0Ω . He has a bandgap reference voltage of 1.0V (can be presented by an ideal voltage source). He has an inductor of $10\mu\text{H}$ and a capacitor of $1.0\mu\text{F}$.

1. Build a voltage regulator to generate the 1.8V supply. Use ideal amplifiers if needed.
 - Calculate and simulate the power efficiency
 - Do you think a voltage regulator is a good choice for this application? Motivate your answer.

2. Build a buck converter to generate the 1.8V supply. Use proper transistors for the power switches. Build a gate driver to drive the power transistors. Use ideal voltage sources to generate PWM control signals. Implement a deadtime of 10ns. PWM switching frequency is 2.0 MHz.
 - Calculate the duty cycle.
 - Calculate the inductor ripple current.
 - Size the power transistors such that the conduction loss is $\sim 5\%$ of the output power. Motivate your choice.
 - Size the gate driver such that the transition time is $\sim 2.0\text{ns}$. Motivate your choice. Check whether there is shoot through.
 - Calculate and simulate the conduction loss with a load of 16Ω .
 - What is the power efficiency @ 16Ω ? What is the dominant power loss?